TEXORA · LECTURE SHEET

Purl Knitting Machine

Purl — links-links — is the only basic structure that changes loop facing course by course. It is knitted with double-headed latch needles that are physically transferred from one bed to the other, a beautiful and slow mechanism that survives where its lengthwise elasticity and true purl texture are worth more than speed.

TE-3101 · KNITTING TECHNOLOGY · TOPIC 7 · LEVEL 3 TERM 1

1. The double-headed needle and the two beds in line

The purl needle has a hook and latch at **both ends** and no butt of its own. It is driven by **sliders (jacks)** that engage a hook at one end and push the needle through the loop-forming positions. The two needle beds lie **in the same plane, facing each other** (flat links-links) or as cylinder/dial equivalents; when the needle is held by the front bed's sliders it knits loops drawn to the front; transferred to the back bed, the other hook knits loops drawn to the back. **Transfer** is the extra machine action: sliders of the receiving bed engage the free hook, the holding sliders release, and the needle changes allegiance — for a whole course at once (plain purl) or needle-by-needle where pattern requires.

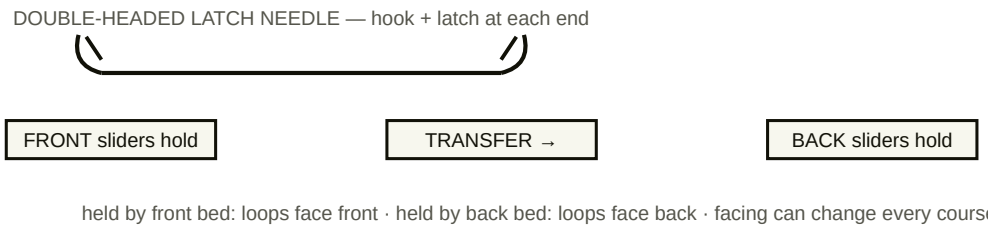


Figure 1 — The double-headed needle and transfer. Whichever bed's sliders hold the needle decides the facing of the next course — the whole principle of links-links.

2. The knitting and transfer cycle

1. **Knit course:** sliders drive all needles through the normal latch cycle (Topic 3) in the holding bed; loops are drawn to that bed's side.
2. **Prepare transfer:** needles are brought to a transfer position with loops on the stems and both hooks accessible.

3. **Engage and release:** the opposite bed's sliders advance, engage the free hooks, and the first bed's sliders withdraw. Needles are now held from the other side. A mis-timed transfer drops or splits loops — the machine's signature fault.
4. **Knit next course** from the new bed: loops drawn the opposite way. Repeat transfer only when the pattern changes facing again — plain 1-course purl transfers every course; garter-type patterns transfer in blocks.

Because transfer occupies machine time in which no loop is formed, purl machines run far below jersey speeds, with few systems; output is measured in panels per hour (knitwear) rather than kg/h of bulk fabric. Loop formation itself is identical in principle — clearing, feeding, knock-over — executed by whichever hook is active, with stitch length set by slider stroke/cam depth as on other machines.

3. Structures and effects

ARRANGEMENT	RESULT	USE
Alternate courses (true purl)	Horizontal ridges, strong lengthwise stretch, balanced	Sweater bodies, scarves
Blocks of courses (e.g. 6 face, 6 reverse)	Garter-like ridged panels	Knitwear design, baby garments (soft ridges, no curl)
Selected-needle transfer (part needles front, part back in same course)	Moss/seed and links patterns combined with rib-type effects	Fashion knitwear

Worked check — lengthwise behaviour: a purl panel knitted at 12 courses/cm on the machine relaxes to 15 courses/cm off the machine (the alternating loop heads pull courses together), a 25 % course-wise contraction — the visible sign of the stored lengthwise elasticity that makes purl garments comfortable and why patterns must be calculated on relaxed, not on-machine, dimensions.

4. Why purl is rare — and where it remains

Costs: slow cycle (transfer time), complex slider mechanisms, double-headed needles are delicate and expensive, gauges are coarse-to-medium, and most purl *appearances* can be imitated on V-bed flat machines by loop transfer stitches. Remaining strongholds: genuine links-links knitwear where both faces are used (reversible garments), traditional sweater production, and as the conceptual parent of transfer patterning on modern flat machines (Topic 10), where the same engage–release–re-engage idea moves loops between beds needle by needle for shaping and cables.

Glossary

Links-links — the trade name for purl knitting (loops linked alternately to each side). **Double-headed needle** — latch needle with a hook at each end. **Slider** — the driver that engages a hook and moves the needle. **Transfer** — handing a needle (or a loop, on flat machines) from one bed to the other.

Key points

• Purl facing is decided by which bed holds the double-headed needle; transfer is an extra, non-knitting action that caps speed.
• Structures: true purl (course alternation), garter blocks, selected-needle moss effects.
• Its living legacy is loop transfer and shaping on V-bed machines.

Practice

1. Explain why a purl machine cannot simply run at rib-machine speed even with perfect mechanics.
2. A designer wants a reversible scarf, identical ridges on both faces, no curl. Choose the structure and transfer programme.
3. Distinguish needle transfer (purl machine) from loop transfer (flat machine) in one paragraph each.
4. A purl panel is 80 cm long on the machine and 64 cm after relaxation. Calculate the course-wise contraction % and state its cause.

Answer key

1. Every facing change consumes a transfer cycle in which sliders engage, release and re-engage all needles without forming loops; at rib speeds the transfer timing window vanishes and dropped/split loops multiply. The limit is the cycle logic, not motor power.
2. True purl, transfer every course: ridges form symmetrically, both faces are the same mixture of face/reverse courses, and the balanced structure lies flat without curl.
3. Needle transfer: the double-headed needle itself changes beds, so the next course is drawn the opposite way; used for course-wise facing changes. Loop transfer: the needle stays in its bed; a formed loop is lifted and placed onto a needle of the opposite bed (or a neighbouring needle), used for shaping, cables and patterns on flat machines.
4. Contraction = (80 − 64) ÷ 80 = **20 %**. Cause: the alternating loop heads and sinker loops pull successive courses together on relaxation — the stored lengthwise elasticity of the purl structure.